TIM 80C, Lecture #11 (5/5/2026)

Agenda :

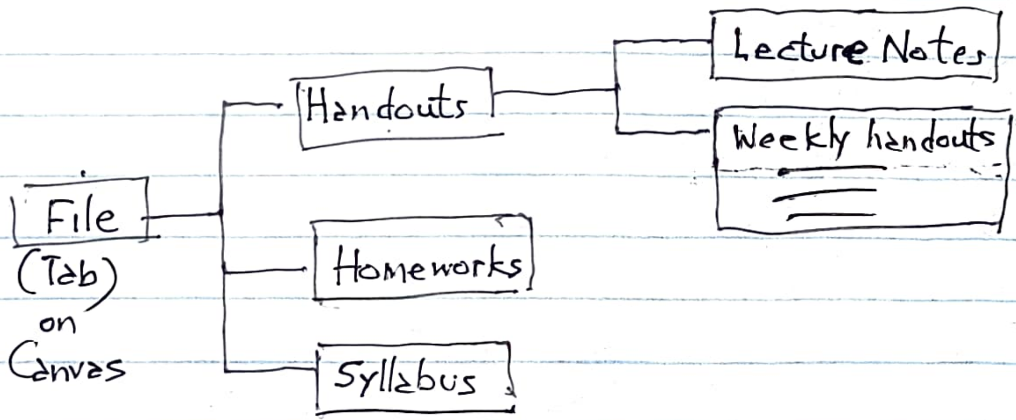
1. Midterm (on Thurday, 5/7/2026 : in-class)

2. Fundamental Financial concepts

3. Cash Flows Analysis for new product development in a start-up

4. HW feedback : very good to excellent!

5. File structure on Canvas



1. Midterm

- Was discussed in L#10
- Complete HW#3 (Mobile-robot startup) before the midterm (10-15 hours)
- The structure of the midterm is the same as the structure of HW#3
- On the midterm do the "Conceptual Product Design" problem last
- Do not rush through the exam
- You will work on the exam document provided in the classroom : paper & pen/pencil
- Open Book, Open Notes, Open Canvas

"Obsidian"

Read your notes

Use your notes

2. Fundamental Financial Concepts

Financial Concepts

(a) Compound Interest :

principal, P, (say \$1,000), placed in a bank account

Annual (compound) interest rate is r (say 10%)

S_1 : money in the account 1 year from now

....

...

S_n : money in your account "n" years from now

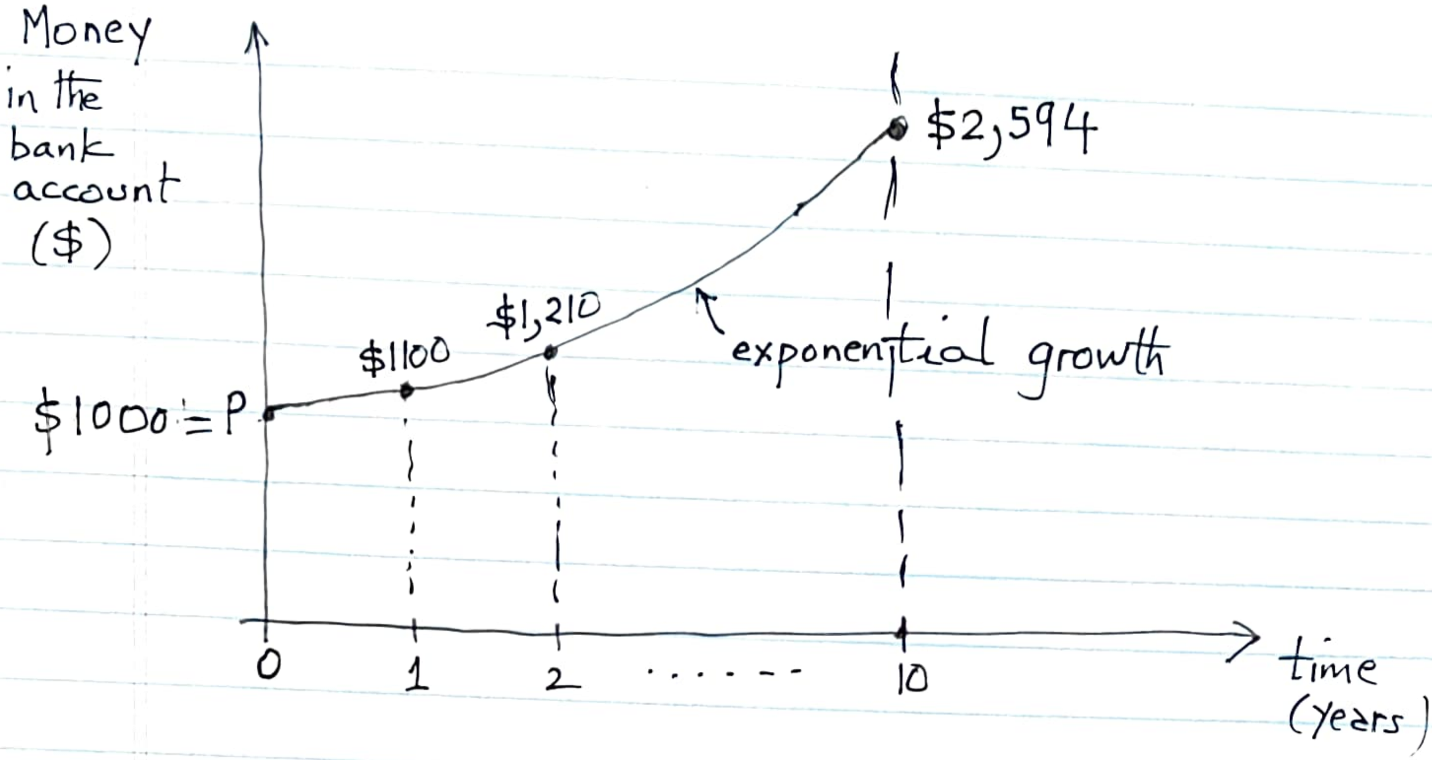
How does compound interest work

$$S_1 = P(1+r) = \$1000(1+0.10) = \$1100$$

$$S_2 = P(1+r)(1+r) = P(1+r)^2 = (\$1000)(1.1)^2 = \$1,210$$

$$\vdots$$

$$S_{10} = \underbrace{P(1+r)(1+r)\cdots(1+r)}_{10 \text{ terms}} = P(1+r)^{10} = \$2,594$$



For a start-up or new product development venture, we are interested in the inverse problem, i.e., the present value (PV) of a projected future cash flow.

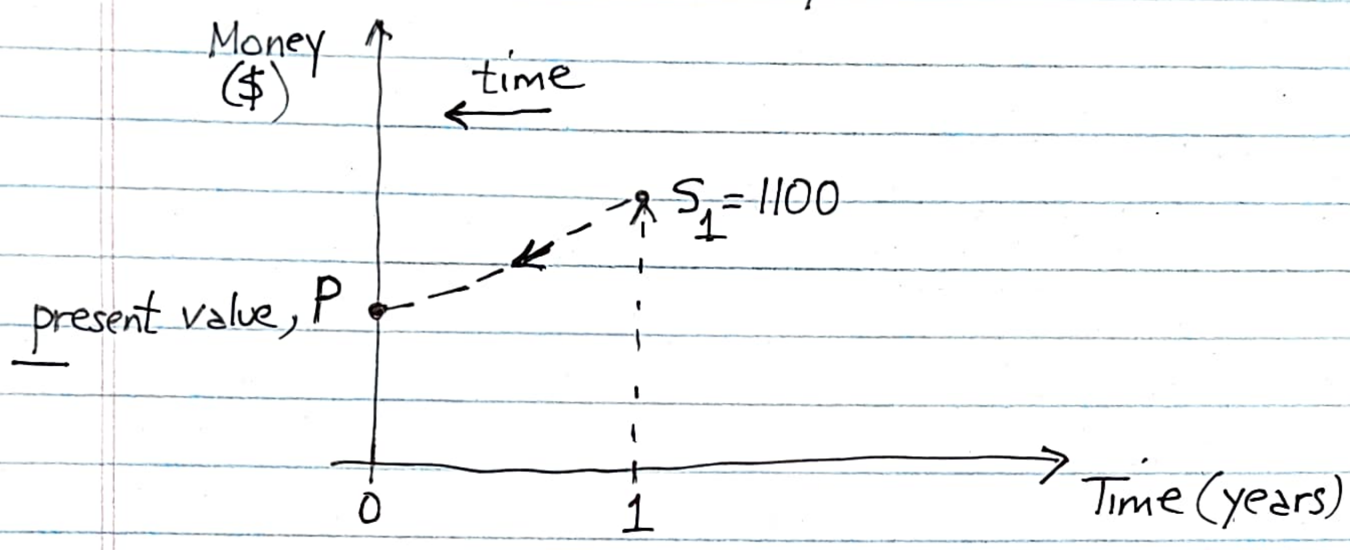
(b) Time Value of Money

Money depreciates with time; i.e. $1 today is worth more (i.e., has more buying power), then $1 a year from today:

$$(\$1)_{\text{today}} > (\$1)_{\text{a year later}}$$

(c) Present Value (PV) of money

Example : If someone promises you a sum of money, S_1 (say \$1100) a year from now, what is S_1 worth today ?



We have to use a discount factor, d, (say 10 %) in order to determine the present value, P, of S_1

$$\text{present value, } P = \frac{S_1}{(1+d)} = \frac{1100}{(1+0.10)} = \$1000$$

(d) Discount factor, d

The discount factor, d, is related to the % cost of borrowing money in order to finance a project

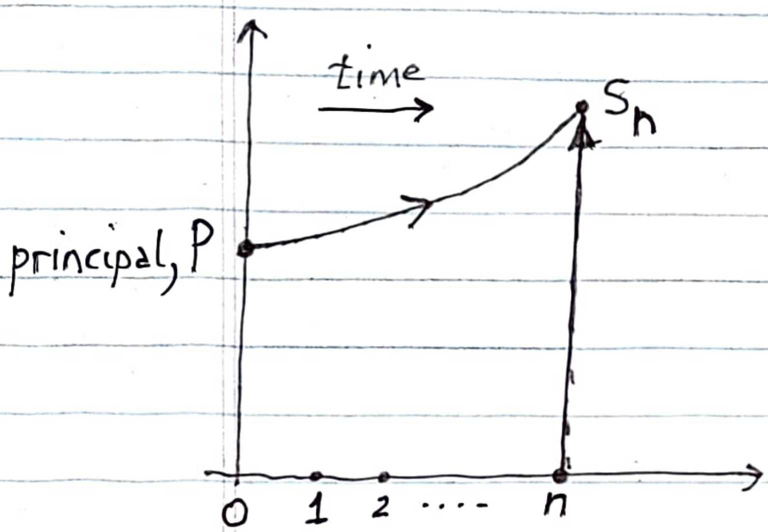
Cost of borrowing money
- Bank : % cost 6% – 12%
- Venture Capitalist : expect an average return of 15% – 25% on their investment.

Discount factor for tech. projects } range from 8% – 15%

(e) Compound Interest

P : principal

S_n : money in the bank account "n" years from now



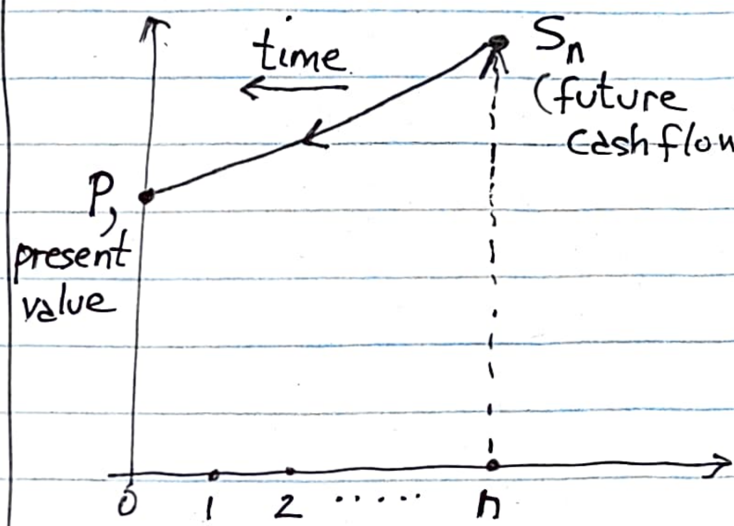
$$S_n = P(1+r)^n$$

r = interest rate

Present Value

P : present value

S_n : projected (or anticipated) future cash flow "n" years from now



$$\text{present value, } P = \frac{S_n}{(1+d)^n}$$

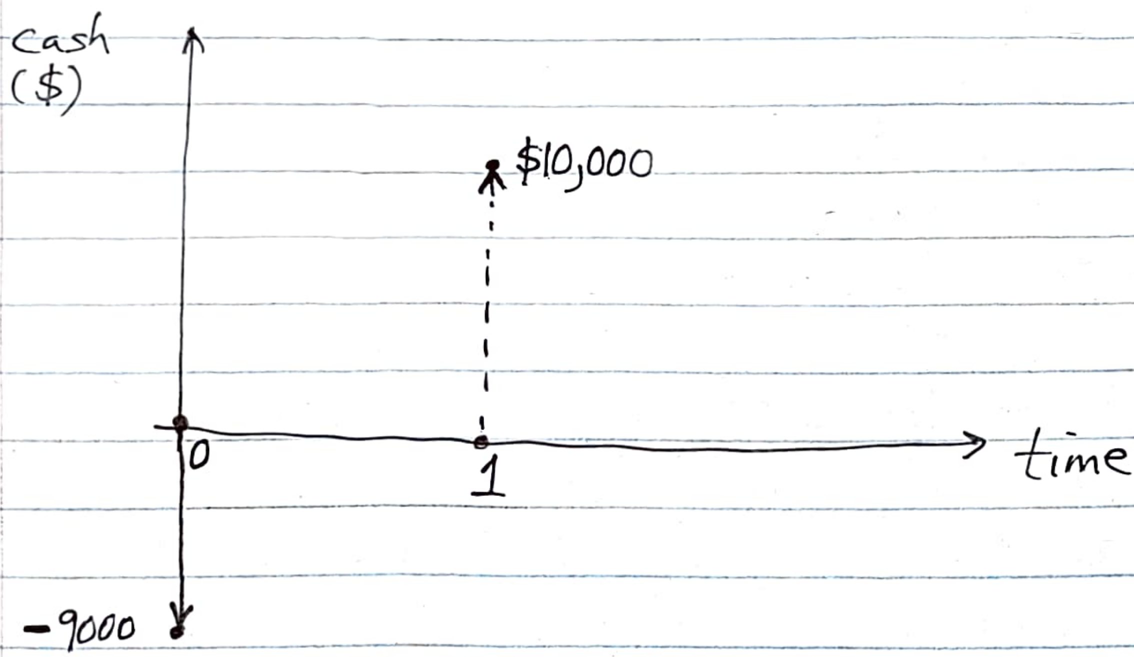
(f) Net Present Value (NPV)

Example

(1) A one period investment

A plot of land is worth \$9,000 today, & is expected to be worth \$10000 a year from now.

What is the present value of a proposed investment, if you purchase the plot of land today, and sell it a year from now.



$$\text{Net Present Value (NPV)} = -9000 + \frac{10,000}{(1+d)}$$

Assume a discount factor, $d = 10\% \equiv 0.10$

$$NPV = -\$9000 + \frac{\$10,000}{1.10} = -9000 + 9090 = \$90 > 0$$

$\Rightarrow$ You may want to invest in buying the plot

Fundamental Financial Principle

- For any project, if the NPV > 0, then go ahead with the project.

$\Rightarrow$ if NPV < 0, then do not perform (or invest in) the project

Application of NPV analysis to a start-up

Problem : What is the expected profit for a new product development project (in a start-up) for a time horizon of "n" years (n = 3 to 5) and an annual discount factor of d %.